Set
Fri, 11 Sep 2026

Problem Set 5 — Proof and disproof

Session 5 · Theorem proving (direct, contrapositive, contradiction, counterexample)

Part A · Reading a statement precisely

1 STATEMENTS

Decide which of these are **statements** (they have a definite truth value) and which are **open sentences**. For each statement, say whether it is true or false and give the reason. For each open sentence, say what you would have to add to turn it into a statement.

a. $x + 1 = 4$.

b. For every real x, $x^2 \geq 0$.

c. There exists an integer n with $n^2 = 9$.

d. There exists a real x with $x^2 = -1$.

2 CONVERSE, CONTRAPOSITIVE

Over the domain $\mathbb{Z}$, consider: *if n is a multiple of 6, then n is even.*

a. Write the converse and the contrapositive in words.

b. Give the truth value of all three statements. Prove or disprove the original and the converse; for the contrapositive, you may quote a theorem from the lecture instead of proving it again.

3 NEGATION

Write the negation of each statement in words, keeping the quantifiers in the right order. Then say which of the two — the statement or its negation — is true, and justify that one.

a. For every real x, $x^2 > x$.

b. There exists an integer n with $n^2 = 5$.

c. For every integer n, if n is prime then n is odd.

4 EMPTY HYPOTHESIS

An implication $P \Rightarrow Q$ is false in one situation only: P true and Q false. So when **nothing satisfies the hypothesis** P, no such situation can arise, there is no case left to check, and the implication is true automatically.

a. Use this to explain, in two sentences, why "every integer n with $n^2 < 0$ is prime" is true.

b. Write a statement of your own about real numbers that is true for the same reason — because no real number satisfies its hypothesis — and say in one sentence why it tells the reader nothing.

Part B · Writing the proof

Name the method at the top of each solution — direct, contrapositive, contradiction or counterexample — then write the proof in full sentences: state the domain, take an **arbitrary** element of it, and finish with the definition the conclusion asks for.

5 DIRECT PROOF

Prove: for all integers m and n, if m and n are odd, then $m + n$ is even.

6 PROVE OR DISPROVE

Each claim below is either true or false. Decide which, then prove it or disprove it with one counterexample. For a counterexample, name the value, confirm it lies in the stated domain, and show exactly what fails.

a. For every real x: if $x^2 > x$, then $x > 1$.

b. For every integer n, $n^2 + n$ is even.

c. For all real a and b: $|a + b| = |a| + |b|$.

7 CONTRAPOSITIVE

Prove: for every integer n, if $3n + 2$ is even, then n is even. Say in one sentence why starting from the hypothesis directly is awkward here.

8 DIRECT · DIVISIBILITY

Prove: for every odd integer n, the number $n^2 - 1$ is divisible by 8. (You may use, with a one-line justification, that the product of two consecutive integers is even.)

9 CONTRADICTION

Prove: if a is rational and b is irrational, then $a + b$ is irrational.

10 UNIQUENESS

Prove that there is **at most one** real number e with the property that $x + e = x$ for every real x. Then name a number with the property, and say what the two facts together establish.

Part C · Reading and writing proofs

11 FIND THE ERROR

A student hands in the following.

> **Claim.** For every $n \in \mathbb{Z}$, if n^2 is even, then n is even.
> **Proof.** Assume n is even. Then $n = 2k$ for some $k \in \mathbb{Z}$, so $n^2 = (2k)^2 = 2(2k^2)$ with $2k^2 \in \mathbb{Z}$, so n^2 is even, which proves the claim. ■

a. Every line of algebra here is correct. Which statement has actually been proved, and what is the name of the mistake?

b. Give a correct proof of the claim.

12 WRITING

In three or four sentences, explain why proving the contrapositive proves the original statement, while proving the converse does not. Include one concrete example of your own.